

Name: Rutuja Waghmare

USN: 2VX24UE043

Expt.No: 4(a)

Title: Floyd's algorithm

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In [2]: def floyd(n,graph):
        for k in range(n):
            for i in range(n):
                for j in range(n):
                    graph[i][j]=min(graph[i][j], graph[i][k]+graph[k][j])
        return graph

n=int(input("Enter the number of vertices: "))
print("Enter the adjacency matrix(enter 'inf' for infinity or '0' for no edge):")

graph=[]
for i in range(n):
    row=[]
    for x in input().split():
        if x.lower()=='inf' or (x=='0' and len(row)!=i):
            row.append(float('inf'))
        else:
            row.append(int(x))
    graph.append(row)

shortest_paths=floyd(n,graph)

print("=====")
print("Shortest paths between all pairs of verices:")
for row in shortest_paths:
    print(*[x if x!=float('inf') else 'inf' for x in row])
print("=====")
```

```
Enter the number of vertices: 4
Enter the adjacency matrix(enter 'inf' for infinity or '0' for no edge):
0 3 inf 7
8 0 2 inf
5 inf 0 1
2 inf inf 0
=====
Shortest paths between all pairs of verices:
0 3 5 6
5 0 2 3
3 6 0 1
2 5 7 0
=====
```